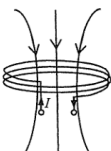
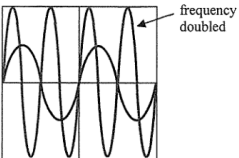


- |    |     |      |                                                                                               |    |          |
|----|-----|------|-----------------------------------------------------------------------------------------------|----|----------|
| 6. | (a) | (i)  | Accelerates at $g$ before the elastic cord stretches / at the beginning.                      | 1A |          |
|    |     |      | Acceleration decreases as the cord stretches.                                                 | 1A |          |
|    |     |      | Decelerates until momentarily at rest                                                         | 1A |          |
|    |     |      | (after the tension in the cord is greater than $mg$ ).                                        |    | <u>3</u> |
|    |     | (ii) | Gravitational potential energy changed to kinetic energy and                                  | 1A |          |
|    |     |      | (then) elastic potential energy in elastic cord.                                              | 1A | <u>2</u> |
|    | (b) |      | Elastic cord lengthens the stopping time,                                                     | 1A |          |
|    |     |      | hence reduces the (net) force acting on the player.                                           | 1A | <u>2</u> |
|    | (c) |      | Contact area is larger,                                                                       | 1A |          |
|    |     |      | hence pressure is smaller during the fall and the structure is less likely to break / detach. | 1A | <u>2</u> |

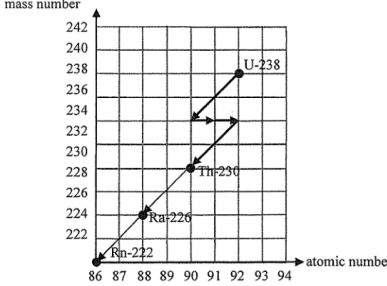
|         |                                                                                                                                                                                                                                                                                                                                                                     |          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10. (a) | Alpha particles emitted can be stopped by the (thin) metallic casing.                                                                                                                                                                                                                                                                                               | 1A       |
|         | <u>Or</u> Shorter range / Lower penetrating power.                                                                                                                                                                                                                                                                                                                  | 1A       |
|         |                                                                                                                                                                                                                                                                                                                                                                     | 1        |
| (b) (i) | $k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{87.74 \times 3.16 \times 10^7}$ $= 2.5 \times 10^{-10} \text{ s}^{-1} \text{ or } 7.9 \times 10^{-3} \text{ year}^{-1}$                                                                                                                                                                                                   | 1M       |
|         | Activity $A = kN$                                                                                                                                                                                                                                                                                                                                                   |          |
|         | $= \frac{\ln 2}{87.74 \times 3.16 \times 10^7} (3.2 \times 10^{25})$ $= 8.000 \times 10^{15} \text{ (Bq)}$                                                                                                                                                                                                                                                          | 1M<br>1A |
|         |                                                                                                                                                                                                                                                                                                                                                                     | 3        |
| (ii)    | Power = Energy per decay $\times$ Activity<br>$= 5.5 \text{ MeV} \times 8.000 \times 10^{15} \text{ Bq}$<br>$= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$<br>$= 7040 \text{ W or } 7.040 \text{ (kW)}$                                                                                                                                | 1M<br>1A |
|         |                                                                                                                                                                                                                                                                                                                                                                     | 2        |
| (iii)   | Power $\propto$ Activity<br>Activity $\propto N$<br>$\therefore$ Percentage of power left $= \left(\frac{1}{2}\right)^{t/t_{1/2}} \times 100\%$<br>$= \left(\frac{1}{2}\right)^{36/87.74} \times 100\%$<br>$= 75.25\% \approx 75\%$                                                                                                                                 | 1M<br>1A |
|         | <u>Or</u> Percentage / fraction of power left = 3/4                                                                                                                                                                                                                                                                                                                 |          |
|         | <div style="border: 1px solid black; padding: 5px;"> <i>Alternatively:</i><br/> <math>N = N_0 e^{-kt}</math><br/> <math>\therefore</math> Percentage of power left <math>= e^{-kt} \times 100\%</math><br/> <math>= e^{-(\ln 2 / 87.74) \times 36} \times 100\%</math><br/> <math>= e^{-0.2844} \times 100\%</math><br/> <math>= 75.25\% \approx 75\%</math> </div> | 1M<br>1A |
|         |                                                                                                                                                                                                                                                                                                                                                                     | 2        |

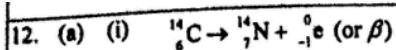
|         |      |                                                                                                                          |                                                                                                                                                            |                         |                                                                                                                                                                                                                                          |         |
|---------|------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 10. (a) | (i)  | $226 - 206 = 20$ (multiple of 4 for $\alpha$ )<br>$\therefore {}^{206}_{82}\text{Pb}$ is the end product                 | 1M                                                                                                                                                         | Accept 97.8 % to 98.0 % |                                                                                                                                                                                                                                          |         |
|         |      |                                                                                                                          | 1A                                                                                                                                                         |                         |                                                                                                                                                                                                                                          |         |
|         |      |                                                                                                                          | 2                                                                                                                                                          |                         |                                                                                                                                                                                                                                          |         |
|         | (ii) | $\% \text{ undecayed Ra-226 left } \frac{N}{N_0} = e^{\frac{\ln 2}{1600} \times 50}$<br>$= 97.857\%$<br>$\approx 97.9\%$ | 1M                                                                                                                                                         |                         |                                                                                                                                                                                                                                          |         |
|         |      |                                                                                                                          | 1A                                                                                                                                                         |                         |                                                                                                                                                                                                                                          |         |
|         |      |                                                                                                                          | <div> <div>Or % undecayed Ra-226 left</div> <div> <math>= \left(\frac{1}{2}\right)^{\frac{50}{1600}}</math><br/> <math>\approx 97.9\%</math> </div> </div> |                         | 1M                                                                                                                                                                                                                                       |         |
|         |      |                                                                                                                          |                                                                                                                                                            |                         | 1A                                                                                                                                                                                                                                       |         |
|         |      |                                                                                                                          | 2                                                                                                                                                          |                         |                                                                                                                                                                                                                                          |         |
|         | (b)  | (i)                                                                                                                      | $\therefore$ random process                                                                                                                                |                         | 1A                                                                                                                                                                                                                                       |         |
|         |      |                                                                                                                          |                                                                                                                                                            |                         | 1                                                                                                                                                                                                                                        |         |
|         |      | (ii)                                                                                                                     | Some of the daughter products of Ra-226 are also radioactive and may emit $\beta$ particles.                                                               |                         | 1A                                                                                                                                                                                                                                       |         |
|         |      |                                                                                                                          |                                                                                                                                                            |                         | 1                                                                                                                                                                                                                                        |         |
|         |      | (iii)                                                                                                                    | Reason: weaker ionizing power of $\beta$ and $\gamma$ radiations                                                                                           |                         | 1A                                                                                                                                                                                                                                       |         |
|         |      |                                                                                                                          |                                                                                                                                                            |                         | <div> <div>- raise the source to a distance greater than the range of <math>\alpha</math> (a few cm) will cease to have sparks.</div> <div>- insert a paper between the source and the gauze, sparks will cease to produce.</div> </div> | Any ONE |
|         |      |                                                                                                                          |                                                                                                                                                            |                         |                                                                                                                                                                                                                                          | 1A      |
|         |      |                                                                                                                          |                                                                                                                                                            |                         | 2                                                                                                                                                                                                                                        |         |

|     |      |                                                                                                                                                                 |                                                                                                                                                                                                                                                       |    |                                                             |
|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------|
| 9.  | (a)  | (i)                                                                                                                                                             | $\beta$ decay / beta decay OR ${}_{19}^{40}\text{K} \rightarrow {}_{20}^{40}\text{Ca} + {}_{-1}^0\beta$                                                                                                                                               | 1A |                                                             |
|     |      |                                                                                                                                                                 |                                                                                                                                                                                                                                                       | 1  |                                                             |
|     |      | (ii)                                                                                                                                                            | Justified: the penetrating power of $\beta$ radiation enables it to penetrate the body's organ / skin<br>Or<br>Not justified: the activity is low and is comparable to background radiation / $\beta$ radiation is largely shielded by the human body | 1A |                                                             |
|     |      |                                                                                                                                                                 |                                                                                                                                                                                                                                                       | 1  |                                                             |
| (b) | (i)  | $\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6}$ (mole)                                                                                                 | 1A                                                                                                                                                                                                                                                    | 1  |                                                             |
|     | (ii) | $k = \frac{\ln 2}{1.25 \times 10^9 (3.16 \times 10^7)} = 1.754803 \times 10^{-17} \text{ (s}^{-1}\text{)}$                                                      | 1M/1A                                                                                                                                                                                                                                                 |    | i. e. $5.545177 \times 10^{-10} \text{ (year}^{-1}\text{)}$ |
|     |      | Activity = $kN$<br>$= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$<br>$= 14.261284 \text{ (Bq)} \approx 14.3 \text{ (Bq)}$ | 1A                                                                                                                                                                                                                                                    | 2  | Accept 14.2 to 14.3 (Bq)                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>12. (a)</p>                                                                                                                                                                                                                                                                                                                                                                     | <p>2A</p>        | <p>1A for correct direction<br/>(also accept field lines pointing upwards for observers with different perception of current direction)<br/>1A for correct field pattern</p> |
| <p>(b) (i) The a.c. flowing in transmitter coil <i>T</i> generates a changing magnetic field, thus by electromagnetic induction an induced e.m.f. is produced in receiver coil <i>R</i> to oppose the changing magnetic flux experienced by it.</p>                                                                                                                                                                                                                 | <p>1A<br/>1A</p> |                                                                                                                                                                              |
| <p>(ii)</p> <p>CRO display</p>                                                                                                                                                                                                                                                                                                                                                     | <p>2A</p>        | <p>1A for correct frequency<br/>1A for correct amplitude</p>                                                                                                                 |
| <p>(c) If the back cover is metallic, eddy currents will be produced (by induction), loss of energy / heating effect results / magnetic field / flux is blocked by the metal case and thus making wireless charging impossible.<br/>Or<br/>Non-metallic materials such as glass are insulators, and no eddy currents are produced.<br/>As a result, energy loss / heating effect will be minimized / the magnetic field can pass through the back cover easily.</p> | <p>1A<br/>1A</p> |                                                                                                                                                                              |

65

| Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks                                                                         | Remarks                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <p>13. (a) (i) <math>k = \frac{\ln 2}{t_{1/2}}</math></p> $= \frac{\ln 2}{3.82 \times 24 \times 3600}$ $= 2.1001405 \times 10^{-6} \text{ s}^{-1} \approx 2.10 \times 10^{-6} \text{ (s}^{-1}\text{)}$ <p>(ii) <math>A = kN</math></p> $N = \frac{48}{2.10 \times 10^{-6}}$ $= 2.285561 \times 10^7 \approx 2.29 \times 10^7$ <p>(iii) As radon is a gas that can be inhaled into the lungs of human, the relatively strong ionizing power of <math>\alpha</math> particles emitted by decaying radon / radioactive gas might affect the organs / cells nearby.</p> | <p>1M<br/>1A</p> <p>2</p> <p>1M<br/>1A</p> <p>2</p> <p>1A<br/>1A</p> <p>2</p> | <p>e.c.f. from (a)(i)<br/>Accept: <math>(2.28 \sim 2.3) \times 10^7</math></p> |
| <p>(b)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>2A</p>                                                                     |                                                                                |

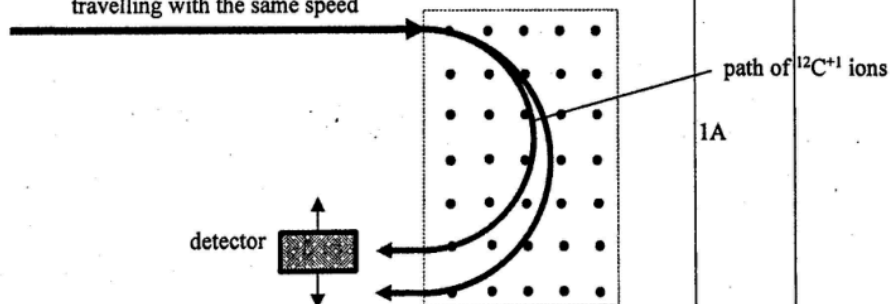


(ii)  $k = \frac{\ln 2}{t_{\frac{1}{2}}}$   
 $= \frac{\ln 2}{5370}$   
 $= 1.2097 \times 10^{-4} \text{ (yr}^{-1}\text{)} \approx 1.21 \times 10^{-4} \text{ (yr}^{-1}\text{)}$

(iii)  $C = C_0 e^{-kt}$   
 $6.0 = 13.6 e^{-(1.21 \times 10^{-4})t}$   
 $t = 6764.7 \approx 6800 \text{ (years)}$

(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.

(ii)  ${}^{12}\text{C}^{+1}$  and  ${}^{14}\text{C}^{+1}$  ions  
travelling with the same speed



(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion, no work is done on the ions.

1A

1

1M

1A

2

1M

1A

e.c.f. from (a)(ii)

Accept: 6760 ~ 6820 (years)

Or

$$(13.6) \left( \frac{1}{2} \right)^n = 6.0$$

$$n \log \left( \frac{1}{2} \right) = \log \left( \frac{6.0}{13.6} \right)$$

$$n = 1.181$$

$$t = n \times 5730 = 6764.7 \text{ (years)}$$

$$\approx 6800 \text{ (years)}$$

2

1A

Or

same number of protons but  
different number of neutrons

1

1A

1

1A

1A

2